

# GAGE WILLIAM BOYD

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## EDUCATION

### Northern Arizona University

Expected May 2028

*B.S. Mathematics & B.S. Computer Science / GPA: 3.6/4.0*

*Flagstaff, AZ*

- **Competitions:** Jump Trading Probability Cup (2026) — **top 220 of 4,012** competitors (top 5.5%)
- **Honors & Activities:** Dean's List, Lumberjack Scholarship, SCLA Honors Society, ACM, FAMUS
- **Coursework:** Statistics, Calculus I–III, Differential Equations, Regression Analysis, OOP

## EXPERIENCE

### Machine Learning Research Assistant

May 2026 – Present

*NAU UQLID Lab — Scientific Machine Learning & Uncertainty Quantification*

*Flagstaff, AZ*

- Benchmarked **11 ML models** (deep ensembles, boosting, Gaussian processes) for bandgap prediction
- Cut bandgap-midpoint RMSE **11.8%** (median error **–25%**) on a held-out 1K set with an MLP ensemble
- Isolated branch-switching at band crossovers as the dominant error; proposed a simulation-side fix

### Project Engineering Intern

Summer 2026

*Achen-Gardner Construction*

*Phoenix, AZ*

- Drafted submittals, purchase orders, subcontracts, and RFIs for a \$16M public-works project
- Cross-checked vendor submittals against materials lists, catching 10+ spec discrepancies
- Prepared Guaranteed Maximum Price (GMP) submittals for a City of Buckeye project

## PROJECTS

### CivilFLOW — Retrieval-Augmented Document Engine | *Next.js, TypeScript, Claude API* | Live Site

2026

- Built a full-stack RAG engine drafting construction purchase orders, subcontracts, and RFIs
- Engineered hybrid BM25 + embedding retrieval, achieving **100% hit@5** and **1.000 MRR**
- Built an eval harness on a 14-case golden set tracking retrieval, citation, and schema checks
- Hardened outputs with JSON schemas, prompt-injection screening, and citation-grounding audits

### GridCast — Probabilistic Electricity-Demand Forecasting | *Python, PyMC, scikit-learn*

2026 – Present

- Ingested **17,500+ hourly observations** of PJM electricity demand from the U.S. EIA API
- Designing calendar, lag, and holiday features plus a seasonal-naïve walk-forward baseline
- Developing Kalman-filter and Bayesian time-series models (PyMC) for calibrated 90% intervals

### Monte Carlo Options Pricer & Backtester | *Python, NumPy, pandas* | GitHub

2026

- Priced European options via 10,000-path Monte Carlo (GBM), validating against Black–Scholes
- Estimated realized volatility from AAPL log returns and plotted the implied-volatility smile
- Backtested a 21-month mean-reversion strategy ( $IV > 1.35x RV$ ), surfacing short-gamma risk

### PlayBook — Sport-Themed Kanban Training Board | *React, TypeScript, Supabase* | Live Site

2026

- Shipped a five-board drag-and-drop Kanban app (dnd-kit, React 19) in a 5-hour take-home window
- Implemented assignees, comments, activity logs, labels, due dates, search/filter, and stats
- Enforced per-user data isolation with Supabase row-level security and anonymous auth

### Full-Stack Retirement Monte Carlo Simulator | *React, FastAPI, NumPy* | Live Site

2026

- Built a 2,000-path geometric Brownian motion simulation engine in Python and NumPy
- Computed median retirement timelines, success rates, and top/bottom-decile outcomes
- Deployed a React/Vite frontend (Vercel) with a FastAPI backend (Render) over a REST API

## TECHNICAL SKILLS

**Programming Languages:** Python, SQL, C++, Java, JavaScript, TypeScript, MATLAB, LaTeX

**Cloud & DevOps:** AWS (EC2, S3, Lambda), Azure, Docker, Linux, PowerShell, Git, Vercel, Render

**ML & Data Science:** PyTorch, scikit-learn, pandas, NumPy, PyMC, statsmodels, Bayesian inference

**AI & Web:** Claude API, RAG, React, Next.js, FastAPI, REST APIs, JSON schema validation